

Marvin Joseph Bogere

AI Engineer | Backend & Production Systems

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PROFESSIONAL SUMMARY

AI and backend engineer building production-oriented systems with Python, FastAPI, LLM APIs, structured outputs, agent orchestration, web search, grounding, and Docker. Brings extensive production SQL Server and data engineering experience across reliability, performance, automation, incident response, and Azure environments. Currently applying that foundation to AI systems designed around validated outputs, provider isolation, source traceability, cost controls, and dependable deployment.

SELECTED AI ENGINEERING PROJECTS

Research Agent - Multi-Agent Research System

Python | FastAPI | Pydantic | asyncio | OpenAI | Tavily | pytest | Docker | Nginx | Ubuntu

- Built a FastAPI workflow that plans one research question into 2-5 focused worker assignments and executes bounded asynchronous research workers.
- Integrated Tavily web search and OpenAI analysis behind application-owned provider interfaces, with runtime-only credentials and mocked automated tests.
- Designed application-owned evidence IDs so models select immutable evidence instead of reproducing or altering factual text.
- Enforced strict claim, evidence, source, and citation grounding with Pydantic validation, deterministic aggregation, explicit conflicts and uncertainties, and partial-worker failure handling.
- Containerized the service for an Ubuntu VPS with Docker, Nginx, HTTPS, CORS restrictions, process-local rate limiting, concurrency controls, and health checks.

Extraction Agent - Multimodal Document Extraction

Python | FastAPI | Pydantic | OpenAI Structured Outputs | pypdf | Vision | pytest | Docker

- Built a FastAPI service for structured invoice extraction from text PDFs, scanned PDFs, and images.
- Routes text PDFs through pypdf and scanned documents or images through vision processing, then validates structured results with Pydantic.
- Added document-grounded follow-up questions and regression coverage for validation, malformed provider responses, uploads, and production failure cases.
- Deployed with Docker, Nginx, HTTPS, environment-based secret management, and health checks on an Ubuntu VPS.

TECHNICAL SKILLS

AI Engineering: LLM APIs, agentic systems, multi-agent orchestration, structured outputs, tool calling, grounding and citations, RAG concepts, prompt engineering, LLM evaluation

Backend Engineering: Python, FastAPI, Pydantic, REST APIs, asyncio, API integration, error handling, pytest

Production & Infrastructure: Docker, Linux, Nginx, VPS deployment, HTTPS, environment and secret management, CORS, rate limiting, health checks, Git, GitHub, CI/CD

Data Engineering: SQL, SQL Server, ETL, Pandas, NumPy, SSIS, CDC, data validation

Production Systems: Monitoring, incident response, performance engineering, HA/DR, automation, reliability engineering, Azure

PROFESSIONAL EXPERIENCE

Diplomatic Solutions Corporation - Senior SQL Server DBA

September 2019 - Present

- Own production SQL Server environments across on-premises and Azure infrastructure, covering 50+ instances, 200+ databases, and multi-terabyte, high-concurrency workloads.
- Support HA/DR, Always On Availability Groups, failovers, backup and recovery, and P1/P2 incident response.
- Diagnose blocking, deadlocks, Query Store behavior, execution plans, waits, I/O, and TempDB performance.
- Automate operational work with PowerShell, T-SQL, and Python where applicable.
- Support Azure migrations and CI/CD database deployment workflows.
- Maintain monitoring, runbooks, recovery practices, and production operational standards.

Emitek - ETL Engineer / SQL Server DBA

July 2018 - August 2019

- Built and supported ETL pipelines across SQL Server, APIs, XML, and flat-file sources.
- Used Python, Pandas, and NumPy for data processing and automation.
- Implemented incremental loads, CDC workflows, monitoring, and data validation.
- Performed SQL query and database performance tuning.

CERTIFICATIONS

Completed

- Microsoft Certified: Azure Database Administrator Associate (DP-300)
- CompTIA Security+

Currently Studying - Not Yet Earned

- Claude Certified Associate - Foundations - Currently Studying
- Claude Certified Developer - Foundations - Currently Studying
- Claude Certified Architect - Foundations - Currently Studying

EDUCATION

Western Governors University

Bachelor of Science in Information Technology - Expected 2026

Montgomery College

Associate Degree - 2021